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Smart Feature Selection for Heart Disease Prediction Using Machine Learning

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Abstract

One of the most common reasons people die is because of heart disease. It is very important to find it early in order to lower the death rate. In the last few years, a lot of people have used machine learning to help them figure out if someone is likely to get heart disease based on their health data. But machine learning models might not work as well when medical datasets have features that aren't useful or are the same as other features.

This project looks at how different ways of choosing features affect how well machine learning can tell if a person has heart disease. Three common ways to pick features are Chi-Square, Mutual Information, and Recursive Feature Elimination (RFE). These techniques help you find the most important parts of the dataset. Then, these chosen features are used to train traditional machine learning models like K-Nearest Neighbours, Support Vector Machine, Random Forest, and Logistic Regression.

The experiments are performed on the UCI Heart Disease dataset, and the performance of the models is measured by accuracy, precision, recall, and F1-score. The outcome of the experiments reveals that feature selection can help in improving the performance of the models and reducing the dimensionality of the features, but the impact of feature selection is different for different models.

Keywords: Heart Disease Prediction, Machine Learning, Feature Selection, Chi-Square, Mutual Information, Recursive Feature Elimination.

Introduction

Heart disease is a major health problem that affects people all over the world. According to reports from around the world, cardiovascular diseases are still one of the main causes of death. This makes early detection and prevention very important. Heart disease often develops without any warning signs, and symptoms may not show up until the disease is very bad. This means that doctors need dependable systems that can help them find patients who are at risk early on.

As digital healthcare data has grown, machine learning has become a useful tool for looking at medical datasets and helping doctors make decisions. Machine learning models can find patterns in old patient data and use them to make predictions about things like age, blood pressure, cholesterol level, and heart rate. These models help people make fewer mistakes and can help doctors by making predictions faster and more reliably.

In medical datasets, though, there are a lot of features, and not all of them are equally important for making predictions. Some features may not be useful or may be the same as others, and using them could make machine learning models less accurate. This problem shows how important it is to choose the right features from a dataset, which is what feature selection is all about.

Feature selection makes models more accurate, stops them from overfitting, and makes the process of making predictions easier to understand. Researchers have come up with a number of ways to choose features in the past. These methods range from simple statistical methods to more complicated optimization algorithms. Even though complicated methods can get very accurate results, they are not easy to understand, especially for research done by students.

The main goal of this project is to do a simple and repeatable analysis of the most common methods for selecting features that are used to predict heart disease. The main goal of this project is to look into how different feature selection methods affect how well traditional machine learning algorithms work. By comparing the performance of models before and after feature selection, this study will elucidate the importance of feature selection in predicting heart disease through machine learning.

The Study's Goals

The main goals of this project are listed below:

- To investigate the utilization of machine learning methods for predicting heart disease.
- To use different methods for selecting features to find the ones that have the biggest impact on predicting heart disease.
- To test and compare how well machine learning models work with and without feature selection.
- To look at how feature selection affects the F1-score, accuracy, precision, and recall of a model.
- To make a machine learning pipeline for predicting heart disease that is easy to use, can be repeated, and is good for students.

Related Work

Several studies have recently examined the application of machine learning techniques for predicting heart disease. Researchers have used different classification algorithms to help with early diagnosis and decision support because medical datasets are now available and data analysis has gotten better.

A lot of early research looked at how well different machine learning classifiers worked with the UCI Heart Disease dataset. Some of these classifiers were Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, and K-Nearest Neighbours. These studies showed that machine learning models can accurately predict heart disease when they are trained on the right clinical features.

Feature selection has become a major focus in models that predict heart disease. To solve the problem of dimensionality, feature selection methods remove features that aren't useful or are too similar to each other. Some writers have used statistical and filter-based methods, like Chi-squared with ensemble and optimization models, to make models that predict heart disease more accurate. These models use hard-to-understand frameworks like boosting, bagging, and evolutionary algorithms. These models are very accurate, but they can also be hard to use and hard to repeat in a simple experiment.

More recent studies have conducted large-scale comparisons of multiple feature selection techniques across different machine learning algorithms to analyze their impact on heart disease prediction performance. These studies highlight that feature selection does not always improve performance uniformly across all models and that the effectiveness of a feature selection technique depends on the classifier used.

In contrast to existing works that emphasize complex or large-scale methods, this study focuses on a simplified and reproducible comparison of commonly used feature selection techniques. By analysing Chi-Square, Mutual Information, and Recursive Feature Elimination with classical machine learning models, this work aims to provide a clear and student-friendly understanding of the role of feature selection in heart disease prediction.

Dataset Description

The heart disease dataset used in this study came from the UCI Machine Learning Repository. This is a well-known and publicly accessible source for machine learning research. Researchers have used this dataset a lot, and it is thought to be reliable for testing models that predict heart disease.

After preprocessing, the dataset has 297 patient records, and each record has information about a patient's health. It has 13 input features that have to do with medical and demographic information, as well as one target variable that shows whether or not heart disease is present. The target variable is binary, meaning that a value of 0 means there is no heart disease and a value of 1 means there is heart disease.

Before doing any experiments, we dealt with missing values in the dataset by deleting records that weren't complete. Then, the dataset was changed into a numerical format that machine learning algorithms could use. These preprocessing steps make sure that the data is consistent and make the experimental results more reliable.

Table 1: Dataset Features

Feature Name	Description
age	Age of the patient
sex	Gender (1 = male, 0 = female)
cp	Chest pain type
trestbps	Resting blood pressure
chol	Serum cholesterol level
fbs	Fasting blood sugar
restecg	Resting ECG results
thalach	Maximum heart rate achieved
exang	Exercise-induced angina
oldpeak	ST depression (exercise)
slope	Slope of peak exercise ST segment
ca	Major vessels by fluoroscopy
thal	Thalassemia
target	Heart disease (0 = no, 1 = yes)

Class Distribution

The dataset has samples from both groups, which are patients with and without heart disease. The class distribution is fairly even, which makes it easier to train

machine learning models without favoring one class too much. The dataset's balanced nature makes it a good choice for testing how feature selection methods affect how well a classifier works.

Methodology

This section explains the overall method used in this project to look at how different feature selection methods affect how machine learning can be used to predict heart disease. This method used is straightforward, well-structured, and easy to replicate which makes it good for research.

Overall Workflow

Figure 1 shows the whole process of the system that was suggested. The first step is to get the dataset and prepare it for use. Next comes selecting features, training the models, and testing them.

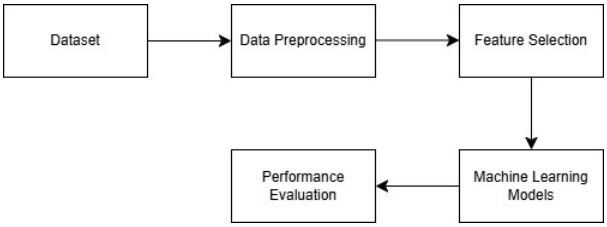

Figure 1: Overall Workflow of the Proposed Methodology

Data Preprocessing

The dataset was cleaned up before machine learning models were used to make sure the data was of good quality and consistent. We got rid of any missing values in the dataset and turned all of the features into numbers. We changed the target variable into a binary format.

Feature Selection Techniques

Feature selection was carried out to determine the most important features that contribute to the prediction of heart disease. In this research work, three popular methods of feature selection were used. For all feature selection methods, the first 8 features were chosen based on ranking score.

Chi-Square: Chi-Square is a method for selecting features based on their statistical relationship with the target variable. Features that have higher Chi-Square scores are thought to be better for classification.

Mutual Information: Another filter-based method is Mutual Information which looks at how much information is shared between a feature and the target variable. It helps find features that are very dependent on the class.

Recursive Feature Elimination (RFE): Recursive Feature Elimination is a method for selecting features that uses wrappers and does so recursively.

Machine Learning Models

To evaluate the impact of feature selection, four classical machine learning models were used: Logistic Regression, Support Vector Machine (SVM), Random Forest, and K-Nearest Neighbours (KNN). These models were selected due to their simplicity, effectiveness, and wide usage in medical prediction tasks.

Experimental Setup

We split the dataset into two parts: a training set and a testing set, with 80% going to the training set and 20% going to the testing set. The Standard Scaler method was used to scale the features. We used accuracy, precision, recall, and F1-score to measure how well the model worked. The full implementation and experimental code for this study is available at: https://colab.research.google.com/drive/124xlmR6hBWbK39Zcdl-hh4fhtcJC_Bvt?usp=sharing

Experimental Results

This part shows the results of the experiments that were done with different machine learning models before and after using feature selection methods. We use accuracy, precision, recall, and F1-score to judge how well the models work. These results help us understand how feature selection changes how well each classifier can make predictions. An 80:20 train-test split was used for all of the tests.

Baseline Model Performance

To begin with, machine learning models were trained on all the available features without using any feature selection method. The outcome of this baseline experiment is presented in Table 2.

Table 2: Baseline Performance of ML Models

Model	Acc.	Prec.	Rec.	F1
Log. Reg.	0.833	0.846	0.786	0.815
SVM	0.850	0.880	0.786	0.830
Rnd. Forest	0.867	0.885	0.821	0.852
KNN	0.883	0.920	0.821	0.868

Performance with Chi-Square Feature Selection

After using the Chi-Square feature selection method, the most relevant features were chosen for training the machine learning models. The performance of the models trained using the Chi-Square selected features is presented in Table 3.

Table 3: Performance with Chi-Square Feature Selection

Model	Acc.	Prec.	Rec.	F1
Log. Reg.	0.867	0.917	0.786	0.846
SVM	0.867	0.955	0.750	0.840
Rnd. Forest	0.817	0.905	0.679	0.776
KNN	0.867	0.917	0.786	0.846

Performance with Mutual Information Feature Selection

Using the Mutual Information feature selection method, the dependency between the features and the target variable was calculated. The selected features were then used for training the machine learning models. The results are presented in Table 4.

Table 4: Performance with Mutual Information Feature Selection

Model	Acc.	Prec.	Rec.	F1
Log. Reg.	0.883	0.920	0.821	0.868
SVM	0.800	0.833	0.714	0.769
Rnd. Forest	0.750	0.783	0.643	0.706
KNN	0.817	0.905	0.679	0.776

Best Performing Model and Performance Improvement

After assessing all the machine learning models based on various feature selection methods, it was noted that the performance of the models differed based on the classifiers and feature selection methods used. Among the models assessed in this research, the combination of Logistic Regression and Recursive Feature Elimination (RFE) performed the best.

Logistic Regression performed well after the application of feature selection methods, which indicates that the model can greatly benefit from the elimination of irrelevant and redundant features. Support Vector Machine also performed well after the application of feature selection methods, but the improvement was relatively lower compared to Logistic Regression. Random Forest performed relatively well with and without feature selection.

Performance Improvement Analysis

To better comprehend the impact of feature selection, the performance of each machine learning model after applying feature selection was compared to its performance without feature selection. Table 5 shows the summary of the performance improvement in terms of F1-score for the best feature selection technique used on each model.

From the above table, it is clear that feature selection impacted the performance of different machine learning models in various ways. The model that experienced the greatest improvement in performance was Logistic Regression, with an improvement of about 6.52% in F1-score after applying Recursive Feature Elimination. Support Vector Machine experienced a slight improvement of about 1.18% when Chi-Square feature selection was applied.

On the other hand, K-Nearest Neighbours and Random Forest showed a drop in performance after feature selection, which could mean that these algorithms are already robust to feature relevance or are sensitive to feature reduction. These findings indicate that feature selection algorithms are more useful for other classifiers and should be selected according to the characteristics of the algorithm.

Table 5: Performance Improvement over Baseline (F1-score)

Model	Base F1	Best F1	Method	Imp.
KNN	0.8679	0.8462	Chi-Sq./MI	−2.50%
Log. Reg.	0.8148	0.8679	RFE	**+6.52%**
Rnd. Forest	0.8519	0.8000	MI	−6.10%
SVM	0.8302	0.8400	Chi-Square	+1.18%

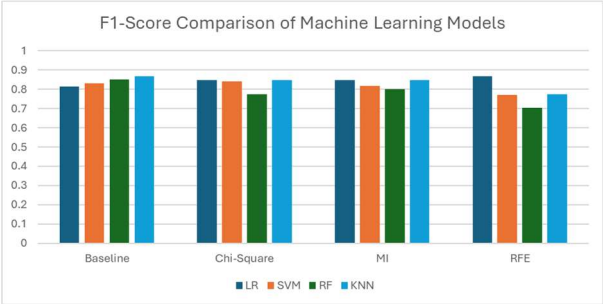

Figure 2: F1-score Comparison Bar Chart

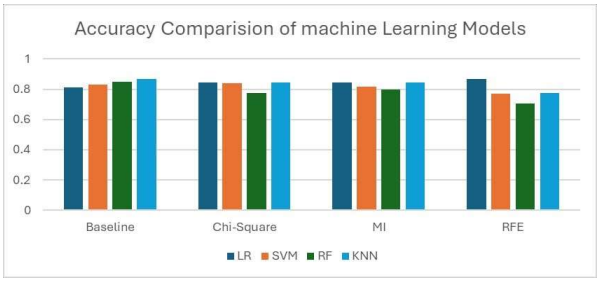

Figure 3: Accuracy Comparison of Machine Learning Models

Discussion

The outcome of this research work clearly shows that feature selection is a significant process in improving the performance of machine learning algorithms in predicting heart disease. Using fewer features as inputs to the machine learning algorithms enables them to focus on the most important features, which can lead to a simpler model with better generalization performance.

From the experimental results, it was noticed that filter-based feature selection algorithms like Chi-Square and Mutual Information achieved moderate levels of improvement in performance for some classifiers. These algorithms are easy to implement and require less computation, making them ideal for student-level and practical scenarios where interpretability is a concern.

The wrapper-based Recursive Feature Elimination (RFE) algorithm showed relatively better performance than filter-based algorithms in some cases. This can be attributed to the reason that RFE takes into account the interaction between the features and the classifier during the selection process. However, RFE also requires more computational resources than filter-based algorithms.

It was also noticed that feature selection does not always result in the improvement of all classifiers. While some classifiers like Logistic Regression and Support Vector Machine benefited from feature selection, tree-based classifiers like Random Forest had limited improvement. This indicates that some classifiers are inherently robust against irrelevant features. From the outcome of this research work, it is clear that the performance of feature selection algorithms depends on the dataset as well as the machine learning algorithm used. Therefore, the selection of a suitable feature selection algorithm depends on the requirements of the problem being considered.

Conclusion

This project gave a comparative study of feature selection methods for the prediction of heart disease using machine learning. The project tried to give a comparison of the effect of Chi-Square, Mutual Information, and Recursive Feature Elimination methods on the conventional machine learning models such as Logistic Regression, Support Vector Machine, Random Forest, and K-Nearest Neighbors.

The experimental result indicated that the use of feature selection methods can help in improving the performance of models by reducing the dimension of the feature space and removing irrelevant features. Among the feature selection methods, Recursive Feature Elimination was comparatively better for some models, whereas filter methods gave easier and better solutions. However, the amount of improvement given by the feature selection methods differed from model to model, which indicates that there is no best method for feature selection for all models.

This project emphasizes the significance of feature selection as a preprocessing step for medical prediction models constructed using machine learning. The result of this project indicates that even the simplest method of feature selection can be employed to construct optimal models for predicting heart disease.

Future Work

This project can be further extended in the following ways. Future work can include the use of other feature selection methods and the use of the proposed method on other medical datasets.

More advanced machine learning and deep learning models can also be used to study their performance on heart disease prediction compared to traditional classifiers. The system can also be tested on larger datasets to improve the robustness of the results.

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